

BusMate

Crowd-Powered, GPS-Free Smart Bus Tracking — Solution Document

Hackathon	CodeX 2026 — MUSA (Maharashtra University Students Association)
Team	Ace Programmers
Domain	Smart Cities & Urban Development
Problem Statement	CX0403 — The Last Mile: Bus Mystery

The Problem

Bus apps often show scheduled timings rather than a bus's real position, especially when official live tracking is unavailable. This hits daily commuters, students, and working professionals hardest — unpredictable arrivals mean longer waits, missed buses, and less confidence in public transport.

Why Existing Solutions Fall Short

- **GPS hardware tracking** (BEST/NMMT trackers): accurate, but needs dedicated devices on every bus — costly to install and maintain.
- **Scheduled-time apps**: show fixed timetables, not real delays or traffic conditions.
- **Raw crowd-sourced tracking** (e.g. generic Moovit-style signals): passenger phone data alone is noisy — people waiting at stops or riding in nearby vehicles get mistaken for the bus.

The gap: no reliable, GPS-independent fallback for live bus tracking.

Our Solution: BusMate

BusMate turns passengers' own smartphones into the sensor network. It anonymously collects location, speed, and direction data, matches movement against known bus routes, and uses **group movement patterns** (several riders moving together along a route) to confidently identify the actual bus — filtering out people who are merely waiting nearby or travelling in other vehicles. The result is a live position and ETA, with no GPS unit ever installed on the bus.

Key Innovation

- **No GPS hardware needed** — passengers' phones act as the sensors.
- **Smart noise filtering** — separates genuine bus passengers from nearby people/vehicles.
- **Crowd-based detection** — clusters multiple riders' signals to confirm a real bus.
- **Smart Signal Trust Score** — rates how reliable each signal is for tracking.
- **Automatic tracking handover** — switches to another reliable signal if the current one drops.

- **Route Chat** — lets riders connect, share updates, and help each other en route.

Target Users

- **Daily bus passengers** — track buses in real time, cut waiting time.
- **Students & working professionals** — reliable arrival info for daily commuting.
- **Public transport authorities** — monitor bus movement, improve routes and service.

How It Works

1. Data Collection	Passengers' phones share anonymous location, speed and direction data.
2. Route Matching	Movement is matched against predefined bus routes to spot potential buses.
3. Noise Filtering	Speed/route-adherence analysis filters out walking, waiting, or nearby-vehicle signals.
4. Bus Detection	Users moving together on the same route are clustered to confirm the actual bus.
5. ETA Prediction	The system calculates current location and estimated arrival time.
6. User Dashboard	Passengers see live location, ETA and route info in the app.

Tech Stack

Flutter / React Native (frontend) • Node.js + Express (backend) • MongoDB / Firebase (database) • Google Maps API (maps) — scalable, cost-effective, real-time.

Feasibility & Expected Impact

Feasibility	Expected Impact
• Uses smartphones — no new bus hardware • Cloud backend — scalable • Testable with simulated passenger data • Accessible development tools • Low battery usage	• More useful live info for passengers • Better tracking during GPS/data gaps • Lower dependence on dedicated hardware • Can support multiple bus operators

Development Plan

Phase 1 — Build	Design a simple passenger-friendly interface; develop backend and real-time location.
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Phase 2 — Crowd Tracking System	Anonymous signal collection; speed/direction & route matching; Smart Signal Trust Score.
Phase 3 — Testing & Integration	Simulate multiple passenger signals; test false signals and automatic handover.
Phase 4 — Final Prototype	Integrate live map + bus location + ETA + confidence score; passenger features; transport-authority monitoring.

Beyond the prototype: AI/ML enhancements (better prediction, advanced noise filtering), multi-modal transport support (metro, auto), city-wide expansion, community features (ratings, feedback), and sustainability goals for greener, smarter public transport.

Solution Summary & Expected Outcome

BusMate is a real-time bus tracking application designed to make public transportation more convenient, reliable, and user-friendly — without requiring any new hardware on buses.

- GPS-independent fallback
- Intelligent signal filtering
- Automatic tracking handover
- ETA with confidence level
- Passenger-friendly participation
- Passenger + transport-authority platform
- Reliable bus-location estimation when live tracking is unavailable
- Reduced uncertainty for passengers
- Better visibility for transport authorities

Team — Ace Programmers

Anshika Singh	Backend & Database Management
Purvesh Koli	Research & Presentation
Adarsh Gupta	UI/UX & Frontend Development
Suryansh Bisen	AI/ML & Data Processing

Original contribution: crowd-powered bus tracking combining live location, route visualization, ETA, confidence level and transport-authority monitoring. Prototype may use public APIs, open-source libraries and datasets, with credits acknowledged.